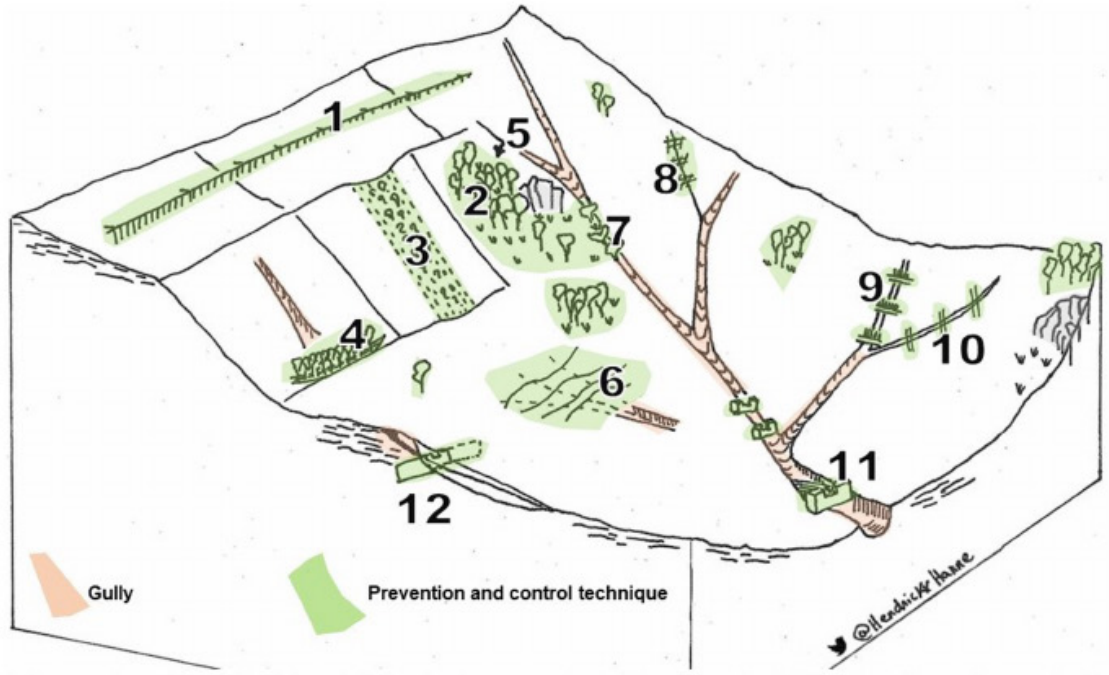


Gully prevention and control technique: 1. Soil and water conservation (soil bund in cropland), 2. Land use change (from cropland to forest), 3. Topsoil resistance in concentrated flow zone (grassed waterway), 4. Vegetation barrier at recurrent ephemeral gully site (fascine made of life vegetation), 5. Runoff diversion into forest, 6. Gully reshaping and filling, 7. Gully channel vegetation, 8. Brushwood check dams, 9. Bioengineered check dams (log dam + live vegetation), 10. Log dams, 11. Loose rock / gabion check dam (showing channel aggradation), 12. Subsurface geomembrane dam to block bypassing of check dams caused by soil piping.

Figure 17: *Overview of gully prevention and control techniques discussed in this paper. Source: Frankl et al., 2021.*

The main outcomes of this paper, summarized by Figure 17, are:

- Preventing gully formation can be done through land use change, applying soil and water conservation techniques or by targeted measures in concentrated flowzones. The latter include measures that increase topsoil resistance and vegetation barriers. Vegetation barriers made of plant residues have the advantage of being immediately effective in protecting against erosion, but have a short life expectancy as compared to barriers made of living vegetation.
- Once deeply incised, the development of gullies may be controlled by diverting runoff away from the channel, but this comes at the risk of relocating the problem. Additional measures such as headcut filling, channel reshaping and headcut armouring can also be applied.
- To control gully channels, multiple studies report on the use of check dams and/or vegetation. Reasons for failures of these techniques depend on runoff and sedi ment characteristics and cross-sectional stability and micro-environment of the gully. In turn, these are controlled by external forcing factors that can be grouped into (i) geomorphology and topography, (ii) climate and (iii) the bio-physical environment.
- The impact of gully prevention and control techniques is addressed, especially regarding their effect on headcut retreat and network development, the trapping of sediment by check dams and reduction of catchment sediment yield. Overall, vegetation establishment in gully channels and catchments plays a key role in gully prevention and control. Once stabilized, gullies may turn into rehabilitated sites of lush vegetation or cropland, making the return on investment to prevent and control gullies high.

Overall, the strategy proposed by CROM DSS is well in line to tackle the problem of the mitigation of gully erosion.